

KAMYSHLOV, Russian Federation

WMO: 284510

Lat: 56.850N

Lon: 62.717E

Elev: 416

StdP: 14.48

Time Zone: 5.00 (E05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.6	-19.5	-29.8	1.1	-24.1	-24.6	1.4	-19.1	18.1	15.5	16.2	14.0	3.1	270	0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	86.1	67.5	82.8	65.9	79.5	64.8	70.3	81.9	68.5	79.1	66.7	76.2	6.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	98.4	74.8	64.6	92.6	73.7	63.0	87.4	72.0	34.4	82.3	33.0	78.9	31.6	76.4	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.7	15.8	13.8	-30.2	91.6	6.4	3.5	-34.8	94.1	-38.5	96.1	-42.1	98.1	-46.8	100.6		
(5)			-30.4	73.0	6.4	2.5	-35.0	74.8	-38.7	76.2	-42.3	77.6	-46.9	79.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	5.2	8.9	23.5	40.2	53.2	62.5	65.2	61.6	50.8	38.2	21.2	9.8
	DBStd	23.76	13.23	13.01	9.01	9.63	9.35	8.12	7.05	7.45	8.41	8.90	13.42	13.81
	HDD50	6350	1389	1152	823	316	77	8	1	3	91	377	865	1248
	HDD65	10513	1854	1572	1288	743	378	139	88	160	433	830	1315	1713
	CDD50	1542	0	0	0	23	177	384	471	361	114	12	0	0
	CDD65	230	0	0	0	0	13	65	93	53	6	0	0	0
	CDH74	2302	0	0	0	8	203	664	900	460	67	0	0	0
	CDH80	633	0	0	0	1	37	192	259	131	13	0	0	0
	WSAvg	5.5	4.9	5.1	6.0	6.5	6.4	5.7	4.7	4.8	5.1	5.8	5.6	5.3
	PrecAvg	19.0	0.9	0.7	0.7	1.0	1.7	2.7	3.1	2.4	1.9	1.5	1.3	1.0
(15) Precipitation	PrecMax	24.4	1.6	1.3	1.8	2.5	4.1	5.4	6.7	5.2	3.6	3.4	2.7	1.9
	PrecMin	12.8	0.1	0.1	0.0	0.1	0.2	1.1	0.6	0.5	0.3	0.1	0.2	0.3
	PrecStd	3.3	0.4	0.3	0.5	0.7	1.0	1.0	1.6	1.1	0.9	0.9	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.8	37.9	46.9	73.5	84.0	89.1	90.6	88.8	82.1	65.9	46.9	37.3
		MCWB	33.0	33.7	38.8	55.6	60.3	68.1	68.5	68.3	61.1	52.9	42.4	34.7
	2%	DB	30.8	34.6	41.8	65.9	79.6	85.4	86.0	83.8	74.3	60.2	42.0	34.2
		MCWB	28.7	31.5	35.4	51.0	59.3	66.0	68.7	67.7	59.1	50.1	39.3	32.1
	5%	DB	27.4	30.9	38.3	60.3	75.1	81.4	82.6	78.9	68.6	55.1	39.2	31.2
		MCWB	25.8	28.1	33.4	48.1	57.3	65.0	68.0	65.4	57.5	47.6	36.8	29.4
	10%	DB	23.9	27.4	35.7	55.3	70.5	77.3	79.1	74.7	63.9	50.7	36.2	27.9
		MCWB	22.4	25.1	31.8	45.4	55.2	63.7	66.4	64.2	55.5	45.4	33.9	26.2
	0.4%	WB	33.3	34.6	39.8	56.0	64.4	71.8	72.4	71.6	64.1	54.2	42.9	34.9
		MCDB	35.4	37.2	45.4	70.7	75.8	83.6	82.9	84.4	73.6	62.8	45.3	36.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.9	31.5	36.5	51.8	61.0	68.9	70.7	69.1	60.8	51.0	39.7	32.2
		MCDB	30.5	34.4	40.7	63.1	75.2	80.7	81.9	80.0	71.1	58.3	41.9	34.1
	5%	WB	25.7	28.5	34.1	49.2	58.8	66.8	69.1	66.9	58.4	48.3	36.9	29.6
		MCDB	27.3	30.7	37.4	59.2	71.5	77.6	80.1	75.7	67.1	54.2	38.9	31.0
	10%	WB	22.6	25.0	32.0	46.3	56.5	64.8	67.5	64.9	56.1	45.7	34.2	26.3
		MCDB	23.9	27.3	35.3	54.7	68.5	75.3	77.2	73.0	62.9	50.4	35.9	27.9
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	15.0	15.2	16.9	20.5	19.2	18.5	17.2	15.7	12.0	10.1	11.4
		MCDBR	11.6	12.1	15.7	24.6	27.6	25.0	24.5	23.2	22.4	20.1	10.8	10.8
		MCWBR	10.7	10.0	11.5	13.5	12.8	10.7	9.9	10.0	11.1	11.9	9.2	9.8
		MCDBR	11.7	10.8	14.0	22.7	24.4	21.9	21.8	20.9	20.3	18.3	10.4	10.6
	5% WB	MCWBR	11.0	9.3	11.1	13.0	12.7	10.7	10.1	10.1	11.2	11.4	9.5	10.1
(40) Clear-Sky Solar Irradiance	taub		0.284	0.315	0.369	0.415	0.413	0.407	0.430	0.441	0.385	0.352	0.310	0.285
	taud		2.044	2.034	1.953	2.082	2.210	2.294	2.229	2.171	2.291	2.254	2.132	2.012
	Ebn at Noon		177	223	240	251	261	265	255	241	239	212	167	138
	Edh at Noon		19	30	43	44	41	38	40	40	31	24	18	15
(44) All-Sky Solar Radiation	RadAvg		226	547	1033	1362	1630	1795	1723	1325	843	428	203	135
	RadStd		20	35	60	133	155	158	141	135	75	71	25	19

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page